

AD 502 – Machine Learning

Pre-Reading Material – Lecture 22

Hierarchical Clustering

Introduction

Hierarchical Clustering is an **unsupervised machine learning technique** used to group similar data objects into clusters. It works with **unlabeled data** and creates a hierarchy of clusters.

The clustering process is commonly represented using a **dendrogram**, which is a tree-like diagram showing how clusters are formed.

Applications

- Customer segmentation
- Document and news grouping
- Image analysis
- Biological data analysis
- Product categorization

Types of Hierarchical Clustering

A. Agglomerative Clustering

Agglomerative clustering follows a **bottom-up approach**.

1. Initially, every data point is considered a separate cluster.
2. The closest clusters are identified.
3. The closest clusters are merged.
4. The process continues until all data points form one cluster.

Example:

A, B, C, D $\rightarrow$ AB, C, D $\rightarrow$ AB, CD $\rightarrow$ ABCD



B. Divisive Clustering

Divisive clustering follows a **top-down approach**.

1. Initially, all data points belong to one cluster.
2. The cluster is divided into smaller clusters.
3. The process continues until the required groups are obtained.

Example:

ABCD $\rightarrow$ AB, CD $\rightarrow$ A, B, C, D

Distance Measurement

Hierarchical clustering uses a **distance or similarity measure** to determine how close or similar two data objects are.

A smaller distance generally indicates that two objects are more similar.

Common distance measures include:

- Euclidean distance
- Manhattan distance
- Cosine distance

Linkage Methods

When clusters contain multiple data points, a **linkage method** is used to determine the distance between clusters.

Linkage Method

Basic Idea

Single Linkage Uses the closest pair of points

Complete Linkage Uses the farthest pair of points

Average Linkage Uses the average distance between points



Ward's Linkage Forms clusters by minimizing the increase in within-cluster variation

Dendrogram

A **dendrogram** is a tree-like diagram that represents the process of forming clusters.

The height at which two clusters are joined represents their **distance or dissimilarity**.

By making a horizontal cut at an appropriate level in the dendrogram, we can determine the **number of clusters**.

Advantages

- The number of clusters does not need to be specified initially.
- Provides a visual representation using a dendrogram.
- Helps understand relationships between data objects.
- Different distance and linkage methods can be used.

Limitations

- Can be computationally expensive for large datasets.
- Results depend on the selected distance and linkage methods.
- Can be affected by noise and outliers.
- Earlier merging decisions are generally difficult to reverse.

Hierarchical Clustering vs. K-Means

Feature	Hierarchical Clustering	K-Means
Learning	Unsupervised	Unsupervised
Number of clusters	Not required initially	Required
Main output	Dendrogram	Cluster assignments
Structure	Hierarchical	Flat

Approach Agglomerative / Divisive Iterative partitioning

Key Takeaways

- Hierarchical clustering is an **unsupervised learning technique**.
- **Agglomerative clustering** follows a bottom-up approach.
- **Divisive clustering** follows a top-down approach.
- A **dendrogram** shows the hierarchy of clusters.
- **Distance measures** determine the similarity between objects.
- **Linkage methods** determine the distance between clusters.
- The number of clusters can be selected by **cutting the dendrogram**.

